

YADHUKRISHNAN PANKAJAKSHAN

krish99.work@gmail.com | [krishpy99.github.io](https://github.com/krishpy99) | [linkedin.com/in/yk2310](https://www.linkedin.com/in/yk2310) | Bay Area, California

EDUCATION

MS in Computer Science | *Northeastern University*

Aug. 2023 – May 2025

Coursework: Algorithms, Software Engineering, Cloud Computing

GPA: 3.92/4.00

EXPERIENCE

Teaching Assistant | Python, C, Java

Jan. 2024 - Present

Northeastern University

- Intro to Data Science - (Fall 2024), Algorithms - (Summer 2024), Programming - (Spring 2024)

Software Engineer | FastAPI, Docker, React

Jun. 2024 - Sep. 2024

Northeastern University

- Worked on Paws, a RAG-based language model for the university with a projected userbase of **4000+** students.
- Integrated **FAISS** for efficient similarity search and retrieval in the context of large-scale document databases.
- Implemented a LlamaIndex retrieval pipeline for prompts, enhancing the overall relevance of responses by **65%**.

Software Engineer | React, Spring, Jenkins

Nov. 2022 – Aug. 2023

Barclays

- Worked with the DevOps & internal applications team and developed **10+** pipelines & features for internal usage.
- Spearheaded the development of **2** REST APIs for Document Management Platform, used by **100k+** customers.
- Wrote auto-scheduled pipelines for reporting that helped generate **6** business reports and saved **\$330** every week.
- Hand-rolled **Logger**, a tool-extension to a Devops web application integrating **z/OS logs** using cURL commands.

Graduate Software Engineer | COBOL, JCL, Sonar, .NET

Sep. 2021 – Oct. 2022

Barclays

- Developed cashback marking algorithms for Barclaycard, delivering **2** mainframe projects within **2 months**.
- Contributed to migration testing team and automated data validation workflows saving **\$950** weekly.
- Innovated data processing algorithms for **100K+** transactions, reducing processing to **linear time complexity**.
- Integration tested developer tools and extensions used by peers, resulting in **12%** increase in peer commit activity.

TECHNICAL SKILLS

C/C++ , Python, C#, Java, JavaScript, .NET, Ruby, Go, Rust, Node, Linux, Docker, Jenkins, Git, AWS, GCP

PROJECTS

ShootClub | C++ , Linux

Sep. 2024 – Present

- Developing a multi-threaded one-player game in C++ utilizing mutex and semaphores for thread synchronization.

FlexiDrop: Reinventing Package Delivery | React, Django, Mapbox

Jun. 2024 – Present

- Integrated Google Maps API for real-time route optimization and live tracking of package delivery.
- Implemented JWT-based authentication to secure user sessions and ensure data integrity.

AutoWrite: A Lazy blogger | NextJS, Flask, LLM

Aug. 2024 - Sep. 2024

- Developed an automated pipeline that generates and auto-publishes blog posts about trending topics on Medium.

The Oracle: A QnA bot | Python, Streamlit

Mar. 2024 - Jun. 2024

- Fine-tuned the deepset/roberta-base-squad2 model on a subset of the MS MARCO dataset (1,000 questions).
- Developed a few-shot query classification system to categorize incoming user queries based on semantic similarity.
- Tested the model using a set of 10K questions achieving good metrics while using (Exact Match) EM and F1 score.

Credit Card Approval Prediction | Python

Jan. 2024 – Apr. 2024

- Built an ML pipeline for credit card approval prediction as part of coursework for DS5110. Trained a baseline **regression** model and compared with other models like **Random Forest** and **XGBoost**.
- Performed hyperparameter tuning using cross-validation to optimize model performance on XGBoost.